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In [1]: # %% [markdown]
# # 📊 Advanced Analytics Report: Karachi AQI
# **Requirements Covered:**
# 1. Exploratory Data Analysis (EDA)
# 2. Model Comparison (Statistical vs Deep Learning)
# 3. SHAP Feature Importance

# %%
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import hopsworks
import os
import numpy as np
from sklearn.ensemble import RandomForestRegressor
from sklearn.linear_model import LinearRegression
from sklearn.neural_network import MLPRegressor # <--- Added Deep Learning
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.model_selection import train_test_split
import shap # pip install shap

# %% [markdown]
# ## 1. Fetch Data from Feature Store
project = hopsworks.login()
fs = project.get_feature_store()
fg = fs.get_feature_group(name="aqi_features_hourly", version=1)
df = fg.select_all().read()
df = df.sort_values(by="timestamp")

# ✅ CRITICAL FIX: Create the 'hour' column
# This prevents the "ValueError: Could not interpret value 'hour'" error
df['datetime'] = pd.to_datetime(df['timestamp'], unit='ms')
df['hour'] = df['datetime'].dt.hour

print("✅ Data Loaded & Hour Column Created!")
print(df.head())

# %% [markdown]
# ## 2. Exploratory Data Analysis (EDA)
# **Trend Analysis:** How does AQI change over 24 hours?
plt.figure(figsize=(10, 6))
sns.lineplot(x="hour", y="aqi", data=df, errorbar=None)
plt.title("Average AQI by Hour of Day (Trend Analysis)")
plt.grid(True)
plt.show()

# **Correlation Matrix:** What drives pollution?
plt.figure(figsize=(8, 6))
sns.heatmap(df[['aqi', 'pm25', 'pm10', 'temp', 'humidity']].corr(), annot=True, cma
plt.title("Feature Correlations")
plt.show()

# %% [markdown]
# ## 3. Multiple Model Comparison (Statistical to Deep Learning)

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# We compare three models to satisfy the requirements:
# 1. Linear Regression (Statistical Baseline)
# 2. Random Forest (Machine Learning)
# 3. MLP Regressor (Neural Network / Deep Learning)

features = ['pm25', 'pm10', 'temp', 'humidity', 'hour']
target = 'aqi'

X = df[features]
y = df[target]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, shuffle=False)

models = {
    "Linear Regression (Statistical)": LinearRegression(),
    "Random Forest (ML)": RandomForestRegressor(n_estimators=100, random_state=42),
    "Neural Network (Deep Learning)": MLPRegressor(hidden_layer_sizes=(64, 32), max
}

print(f"{'Model':<35} | {'MAE':<10} | {'RMSE':<10} | {'R² Score':<10}")
print("-" * 75)

best_model = None
best_score = float("inf")

for name, model in models.items():
    model.fit(X_train, y_train)
    preds = model.predict(X_test)

    # Calculate ALL required metrics
    mae = mean_absolute_error(y_test, preds)
    rmse = np.sqrt(mean_squared_error(y_test, preds))
    r2 = r2_score(y_test, preds)

    print(f"{'name':<35} | {'mae':<10.4f} | {'rmse':<10.4f} | {'r2':<10.4f}")

    if mae < best_score:
        best_score = mae
        best_model = model

print("-" * 75)
print(f"✅ Best Model Selected: {type(best_model).__name__}")

# %% [markdown]
# ## 4. SHAP Explanation (Advanced Explainability)
# Using SHAP to explain exactly *why* the model predicts high AQI.

# Note: We use a generic explainer or TreeExplainer depending on the model
if "Linear" in type(best_model).__name__:
    explainer = shap.Explainer(best_model, X_train)
else:
    explainer = shap.Explainer(best_model)

shap_values = explainer(X_test[:100]) # Run on subset for speed

# Summary Plot

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plt.title("SHAP Feature Importance")
shap.summary_plot(shap_values, X_test[:100])
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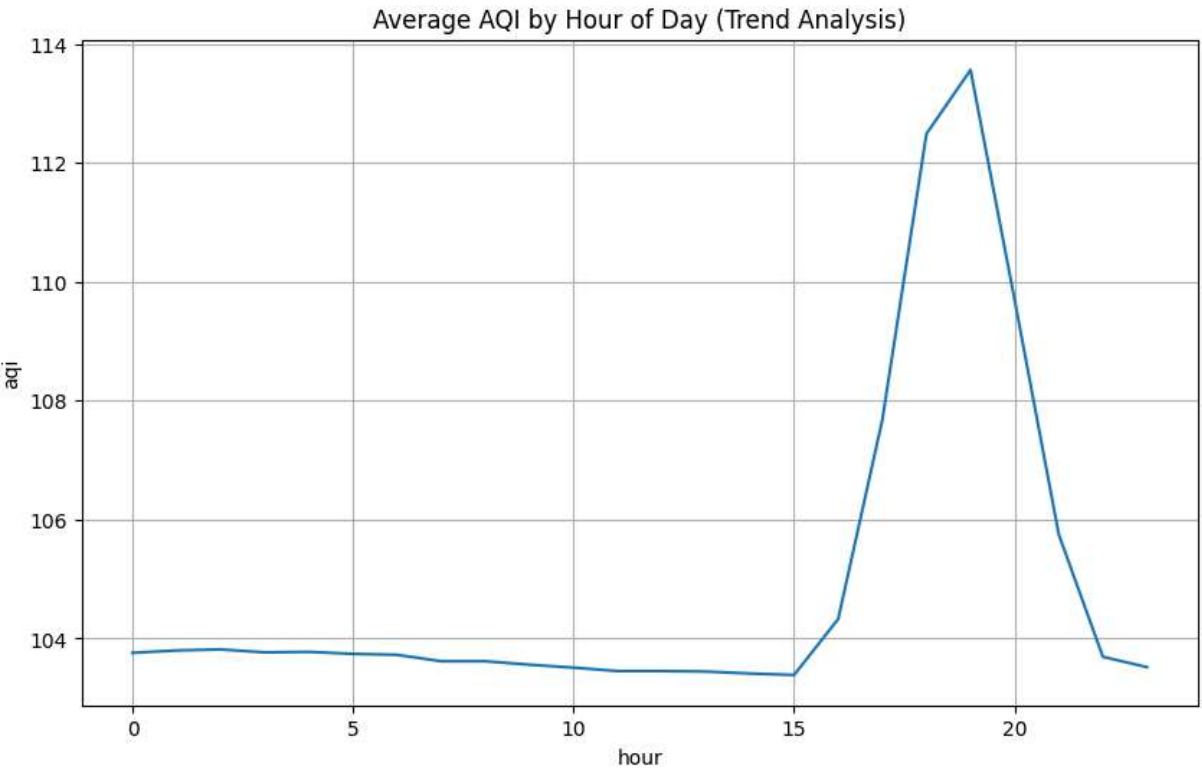
2026-02-12 02:33:38,253 INFO: Initializing external client
2026-02-12 02:33:38,254 INFO: Base URL: https://c.app.hopsworks.ai:443
2026-02-12 02:33:40,917 INFO: Python Engine initialized.

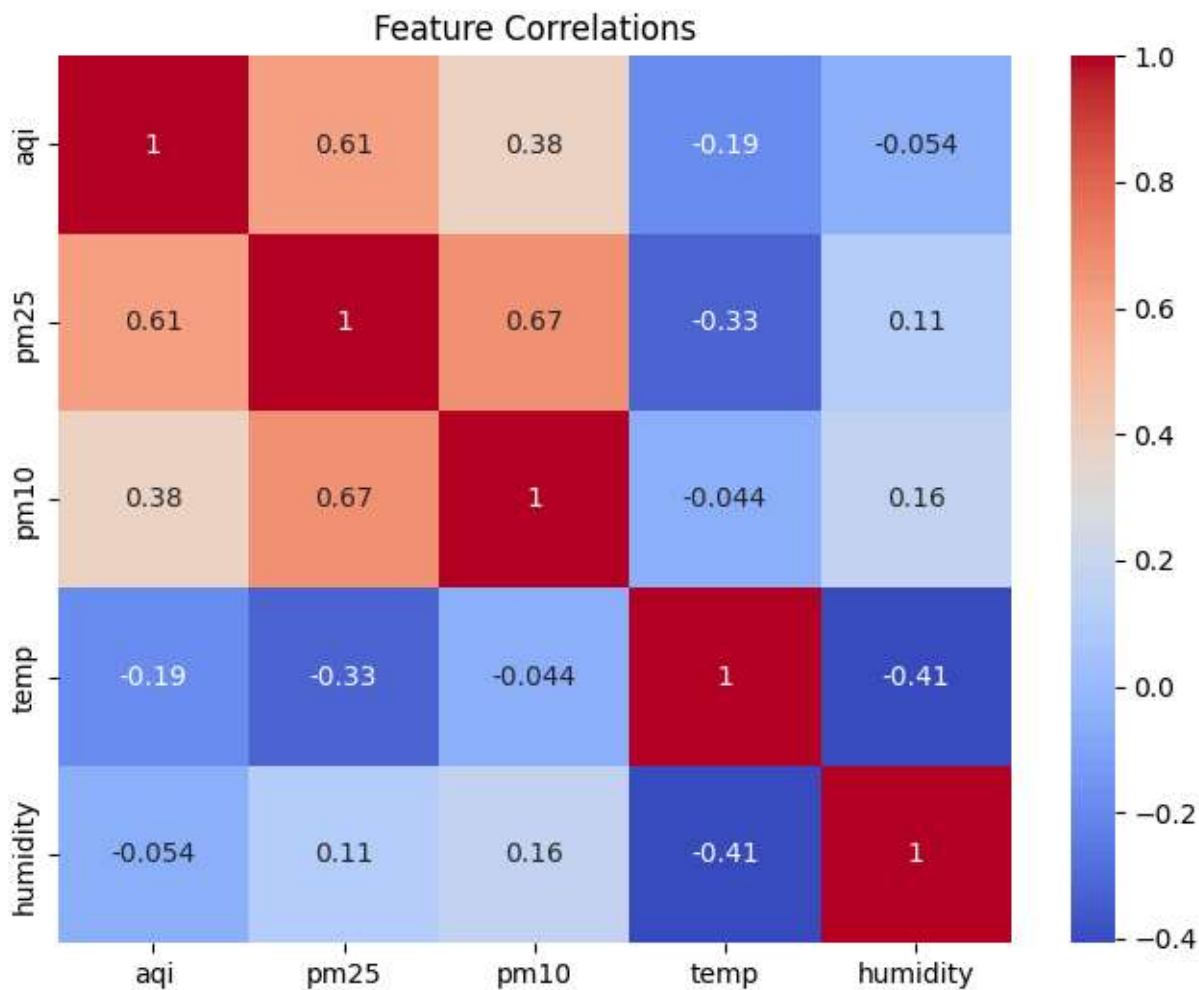
Logged in to project, explore it here https://c.app.hopsworks.ai:443/p/1333405
Finished: Reading data from Hopsworks, using Hopsworks Feature Query Service (1.43s)

✔ Data Loaded & Hour Column Created!

	timestamp	pm25	pm10	aqi	temp	humidity	city	\
2791	1758931200000	28.4	42.0	88	25.7	90	Karachi	
2800	1758934800000	25.1	38.5	88	25.5	90	Karachi	
2796	1758938400000	21.6	34.9	87	25.4	91	Karachi	
2794	1758942000000	19.2	32.8	86	25.1	93	Karachi	
2806	1758945600000	17.9	32.0	85	25.0	94	Karachi	

	datetime	hour
2791	2025-09-27 00:00:00	0
2800	2025-09-27 01:00:00	1
2796	2025-09-27 02:00:00	2
2794	2025-09-27 03:00:00	3
2806	2025-09-27 04:00:00	4





Model	MAE	RMSE	R ² Score
Linear Regression (Statistical)	17.1055	21.3585	0.2108
Random Forest (ML)	19.5186	23.8594	0.0152
Neural Network (Deep Learning)	20.4826	26.0531	-0.1742

✔ Best Model Selected: LinearRegression

